

Dataset alignment, execution and scientific evaluation

P08 Observed-output schema extraction

SYSTEM	Infer id, sequence and probability fields only from visible output headers and samples. Never invent a column. If an identifier or prediction field cannot be established, return UNKNOWN and request human intervention.
USER	Read the Stage-1 exploration report and return the exact seven-key schema for every model: file_path, file_ext, sep, comment_char, id_col, seq_col and prob_col. Output JSON only.
VERIFIED OUTPUT	Observed schemas are persisted and reused through schema_memory.json
SOURCE	agents/runtime_prompts/data_analyst_extraction.md · run_meeting.py:330-331

P09 Evaluation code generation and review

SYSTEM	Use only runtime-supplied model commands, dataset files, schemas and output contracts. Do not invent paths, dependency versions, model outputs, columns or metric values. Preserve missing predictions as missing.
USER	Generate one executable Python evaluation script and one launch script. A separate Data Architect reviews every referenced field; runtime code performs syntax, artifact, dependency, path and scientific metric validation.
VERIFIED OUTPUT	Automatic HPC predictions and manual prediction tables converge at one evaluator
SOURCE	agents/runtime_prompts/coder.md · data_analyst_review.md · main.py · run_meeting.py

P10 Manual prediction import contract

SYSTEM	Manual model outputs cannot bypass provenance or metric rules. Preserve the original file, align sequence IDs, labels and probabilities, and report duplicate, unmatched or invalid rows.
USER	Import the three supplied prediction tables into dataset-specific directories, standardize their schema and continue from the same scientific evaluator used by automatically executed models.
VERIFIED OUTPUT	C_AMPs-predict test n=59,311 · Veltri test n=1,203 · ProteoGPT test n=1,796
SOURCE	import_manual_prediction_results.py · data/manual_predictions/ · data/results_manual/

P11 Independent scientific result audit

SYSTEM	Interpret objective benchmark data without fabricating missing results. Distinguish execution failure from biological prediction performance and keep dataset-specific limitations visible.
USER	Review dynamic metric weights, real metric tables and quantitative scores. Explain performance under the meeting protocol, then write the model-level scientific judgement and evidence-limited ranking.
VERIFIED OUTPUT	Three evaluation bundles produced for 18 usable models
SOURCE	agents/runtime_prompts/critic.md · scientific_evaluation.py · data/results_manual/*/eval_result.json